

21	The charge of a uranium nucleus is $1.5 \times 10^{-17} \text{ C}$. What is the potential at the mid-point between a uranium nucleus and an alpha particle if they are separated by a distance of $1.0 \times 10^{-13} \text{ m}$?	
	A	$2.2 \times 10^{-13} \text{ V}$
	B	$4.3 \times 10^{-13} \text{ V}$
	C	$2.8 \times 10^6 \text{ V}$
	D	$5.5 \times 10^6 \text{ V}$

Answer: C

Alpha particle is a ${}^4_2\text{He}$ nucleus, hence its charge = $+2e = 3.2 \times 10^{-19} \text{ C}$

$$\text{Using } V = \frac{Q_1}{4\pi\epsilon_0 r_1} + \frac{Q_2}{4\pi\epsilon_0 r_2}$$

Since the distances from the two charges to mid-point are the same, $r_1 = r_2$.

$$V = \frac{1.5 \times 10^{-17} + 3.2 \times 10^{-19}}{4\pi(8.85 \times 10^{-12})(0.5 \times 10^{-13})} = 2.8 \times 10^6 \text{ V (2 s.f.)}$$